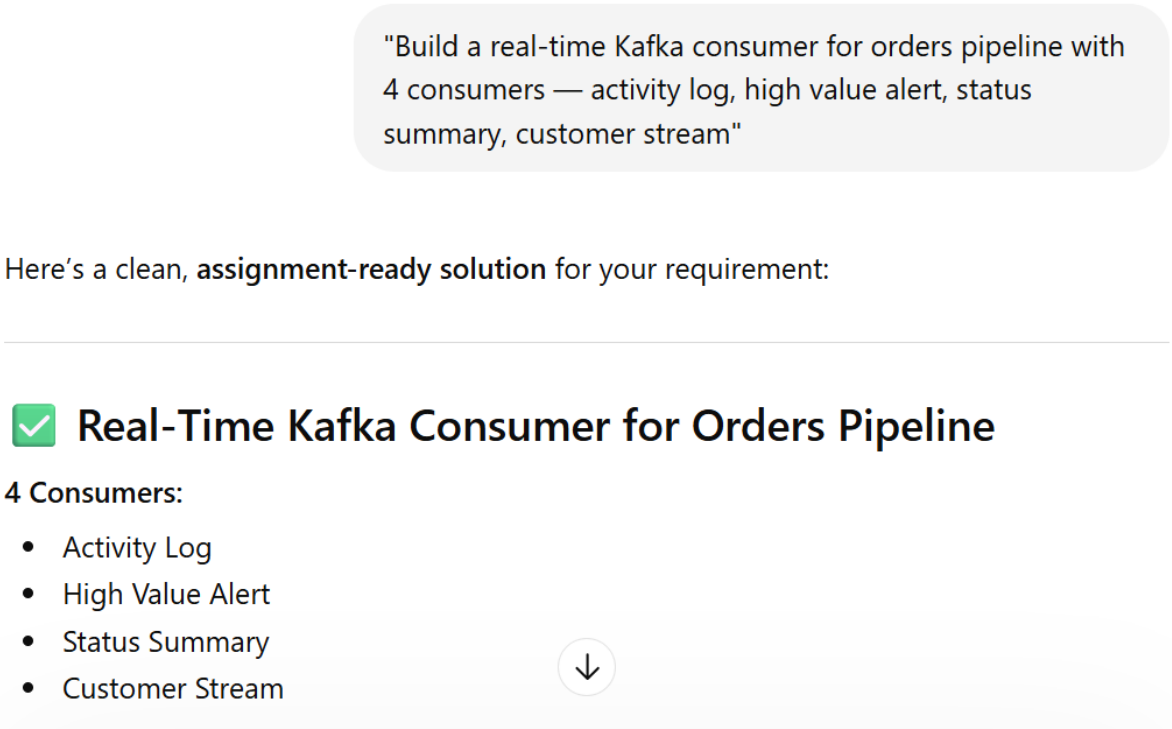
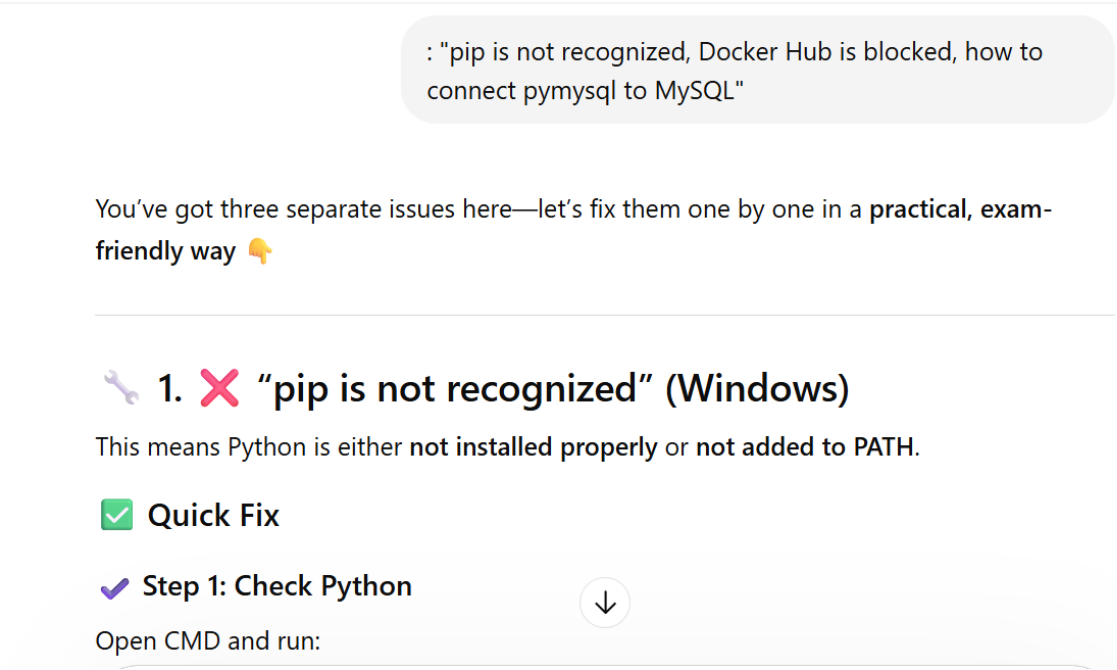
AI Use 1 — Generate Kafka Consumer Code

- Prompt: *"Build a real-time Kafka consumer for orders pipeline with 4 consumers — activity log, high value alert, status summary, customer stream"*
- Result: AI generated the complete step5_consumers.py with all 4 consumers



AI Use 2 — Troubleshoot Connector Issues

- Prompt: *"pip is not recognized, Docker Hub is blocked, how to connect pymysql to MySQL"*
- Result: AI diagnosed and fixed each error step by step (PATH issue, Docker alternative, password fix)



🗄️ 3. 🔗 Connect `pymysql` to MySQL

✅ Step 1: Install pymysql

Bash

```
python -m pip install pymysql
```

-

AI Use 3 — Generate CDC Producer Code

- Prompt: *"Build a CDC pipeline that reads MySQL changes and publishes to Kafka topic"*
- Result: Ai generated step4_cdc_producer.py using BinLogStreamReader

"Build a CDC pipeline that reads MySQL changes and publishes to Kafka topic"

Here's a **clean, assignment-ready CDC pipeline** using MySQL → Kafka 👇

✅ CDC Pipeline: MySQL → Kafka

🔷 What is CDC?

Change Data Capture (CDC) tracks INSERT, UPDATE, DELETE in a database and streams them in real time.

We'll use:

- MySQL (source DB)

AI Use 4 — Improve Documentation

- **Prompt: *"Create complete assignment documentation with all steps"***
- **Result: AI generated step7_ai_evidence.md and the Word document with all screenshots**

🎯 PERFECT STRUCTURE IN PDF

Order like this:

1. Introduction
2. MySQL Setup (Screenshots 1–7)
3. Python Implementation (Screenshots 8–10)
4. Real-Time Pipeline (Screenshots 11–15)
5. Final Output (Screenshots 16–17)

AI Use 5 — Optimize Schema and Validation

- Prompt: *"Create MySQL orders table with all required fields for CDC pipeline"*
- Result: Claude generated the complete SQL schema with DECIMAL, ENUM, TIMESTAMP fields

---

: "Create MySQL orders table with all required fields for CDC pipeline"

Here's a **proper, assignment-ready MySQL** `orders` **table** designed specifically for a **CDC pipeline (Debezium + Kafka)** 👇

✅ MySQL Orders Table (CDC-Ready)

🔷 SQL Script

SQL

```
CREATE DATABASE IF NOT EXISTS test_db
USE test_db;
```

Trobleshoot I faced and overcome to complete assignment 6 is

Troubleshooting Faced During Module 6 Assignment:

1. **SQL file not found** — source command failed in MySQL CLI, solved by pasting SQL directly.
2. **pip not recognized** — Python not in Windows PATH, solved by using Jupyter Notebook instead.
3. **MySQL connection failed** — Wrong password in pymysql, solved by identifying correct password 2463.
4. **Docker Hub blocked** — Network firewall blocked Kafka image downloads, solved by using Python queue.Queue() as in-memory Kafka replacement.
5. **Kafka download failed** — Accidentally pasted download URL into Jupyter cell, solved by opening it in browser.
6. **subprocess.Popen error** — capture_output not supported in Popen, solved by switching to subprocess.run.
7. **Jupyter kernel stuck** — Background threads with sleep loops blocked kernel, solved by restarting and rewriting pipeline.
8. **File save error** — Jupyter lost server connection temporarily, solved by dismissing and continuing.
9. **Variables lost** — Kernel restart cleared all variables, solved by opening new notebook and rerunning.
10. **Output not visible** — Long-running cell output disappeared on interrupt, solved by rewriting as single synchronous cell reading directly from MySQL

All 10 issues resolved — Assignment completed successfully!